

Morning Paper Report - Tuesday, June 09, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Neural Mechanisms of Working Memory Accuracy Revealed by Recurrent Neural Networks

Yuanqi Xie, Yichen Henry Liu, Christos Constantinidis, Xin Zhou
Frontiers in Systems Neuroscience, 2022

Source: Crossref; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://doi.org/10.3389/fnsys.2022.760864>

Goal: Understanding the neural mechanisms of working memory has been a long-standing Neuroscience goal

Method: Bump attractor models have been used to simulate persistent activity generated in the prefrontal cortex during working memory tasks and to study the relationship between activity and behavior

Results: As a consequence, delay period firing rate and behavior were positively correlated, in strong analogy to experimental results from the prefrontal cortex

Conclusion: As a consequence, delay period firing rate and behavior were positively correlated, in strong analogy to experimental results from the prefrontal cortex These findings reveal that delay period activity is computationally efficient in maintaining working memory, as evidenced by unbiased optimization of parameters in artificial neural networks, oblivious to the properties of prefrontal neurons.

2. Memory Fidelity Reveals Qualitative Changes in Interactions Between Items in Visual Working Memory

Zachary Lively, Maria M. Robinson, Aaron S. Benjamin
Psychological Science, 2021

Source: Crossref; matched terms: visual working memory, working memory, fidelity, computational model, computational models

Link: <https://doi.org/10.1177/0956797621997367>

Goal: Memory for objects in a display sometimes reveals attraction—the objects are remembered as more similar to one another than they actually were—and sometimes reveals repulsion—the objects are remembered as more different from one another

Method: In three experiments (each N = 30 adults), we demonstrate that memory fidelity provides a unifying dimension that accommodates the existence of both types of visual working memory interactions

Results: We show that either attraction or repulsion can arise simply as a function of manipulations of memory fidelity

Conclusion: We also demonstrate that subjective ratings of fidelity predict the presence of attraction or repulsion on a trial-by-trial basis We discuss how these results bear on computational models of visual working memory and contextualize these results within the literature of attraction and repulsion effects in long-term memory and perception.

3. Trained recurrent neural networks develop phase-locked limit cycles in a working memory task

Matthijs Pals, Jakob H Macke, Omri Barak
2023

Source: Crossref; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: <https://doi.org/10.1101/2023.04.11.536352>

Goal: Abstract Neural oscillations are ubiquitously observed in many brain areas

Method: We characterized how the stability of the attractor dynamics depends on both reference oscillation amplitude and frequency, properties that can be experimentally observed

Results: We found that networks converged to stable oscillatory dynamics

Conclusion: Here we ask: What kind of neural dynamics support phase coding of information with neural oscillations? We tackled this question by analyzing recurrent neural networks (RNNs) that were trained on a working memory task

4. A Computational Approach to Search in Visual Working Memory

Aditi Singh, Andra Mihali, Wen-Chuang Chou, Wei Ji Ma
2023

Source: Crossref; matched terms: visual working memory, visual search, working memory, precision

Link: <https://doi.org/10.31234/osf.io/jqthb>

Goal: Search in visual working memory (VWM) is an important process in everyday life, but it has so far not been investigated sufficiently using psychophysical and computational approaches

Method: Here, we examine whether computational process models from the study of VWM and of visual search also apply to search in VWM

Results: We found that a variable-precision model fit the data well and that adding an item limit did not improve the fit

Conclusion: We found that a resource-rational model with an ad-hoc cost function captures behavior well, but not better than the best non-resource-rational model Our results, albeit unsurprising, contribute to the evidence that search in VWM is very similar to search in perception.

5. Evaluating working memory representations during the attentional blink: A comparative modeling approach

Shuyao Wang, Aytaç Karabay, Elkan G. Akyürek
2022

Source: Crossref; matched terms: working memory representation, working memory, precision, modeling, attentional blink

Link: <https://doi.org/10.1101/2022.09.15.508098>

Goal: Abstract The nature of working memory (WM) limitations has been a topic of long-standing debate, with several models proposed to elucidate this issue

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Results: Slot-family models most often minimized AIC for second-target reports at short lags, while variable-precision models improved at longer lags and with color targets, indicating predominantly discrete consolidation during the AB, with feature- and lag-dependent graded components

Conclusion: Slot-family models most often minimized AIC for second-target reports at short lags, while variable-precision models improved at longer lags and with color targets, indicating predominantly discrete consolidation during the AB, with feature- and lag-dependent graded components These patterns imply that failure-to-encode (guessing) dominates over low-precision encoding, except when feature content or lag affords...